8D report

Customer plant	Eurotech	**Supplier**	AVO Carbon Germany	**Date of claim**	22.02.2024
Customer contact	A.Strnatkova	**AVO contact**	Joachim Gehl		
Customer reference	24010	**AVO reference**	240203		

QUICK RESPONSE (D1-D3)
This document must be issued within 24 hours after claim notification

D 1. Description of the problem

1.a What is the problem ? Use 5W+2H to describe what is the current situation.

Problem description	Damaged wire on the carbonbrush, flag burr > 2mm, soft compaction	What happened?	Damaged wire on the carbonbrush, flag burr > 2mm, soft compaction
Incident type	Customer claim	Why is it a problem?	Parts out of spec.Can not be assembled
PN customer/AVO	3 134 320 622/ 1026151	When did it happen?	No information
Part description	Carbon brush	Who detected it?	No information
AVO batch number	276542, 277588	Where was it detected?	No information
Recurrence	non reccurent	How was it detected?	No information
Defect S/R	no	How many nOK parts ?	43 NOK assembled+ 5 NOK single
Standard process used	yes		

nok part picture / sketch

ok part picture / sketch

1.b Problem solving team

Name	Position	Location	Email
Joachim Gehl	QM	AVO Germany	joachim.gehl@avocarbon.com
Martin Röhr	PRD	AVO Germany	
Mustafa Yorulmaz	KVP	AVO Germany	
Michael Ott	QM	AVO Germany	
Giovanni Balenzano	PRD	AVO Germany	

1.c Tools to use

yes	Ishikawa		5why's		LLC		Audit		Other

1.d First analyse of problem / potential root causes

Potential root cause(s):

The samples are mandatory for the deeper analysis. Feel free to use our FedEx account. (Please see below marked in yellow)

Samples of nok parts required in plant for analyse: yes

Express shipping number account: FedEx 186141300

D 2. Risk on similar products and processes

Can this defect happen on similar parts ?

		Comments (compulsory if yes)
Other family product	No	Possible on all welding units
Other line	No	
Other tools/work station	No	
Other	Yes	

D 3. Containment

Sorting criterias (add additional sheets if needed)

Location in the supply chain	Nb of involved parts	Nb parts sorted	Nb nok parts	ppm	Date	Resp.
Parts at customer	**170.000**	---	**48**	---	23.02.2024	A.Strnatkova
Parts in transport		---	---	---	---	---
Parts at AVO Germany stock		---	---	---	---	---
Parts at AVO production	---	---	---	---	---	---

First contained shipment information

Immediate actions to prevent production and delivery of nok parts ?

Actions	Resp.	Date plan.	Date clos.
Register Complaint	G. Luschtinetz	23.02.2024	23.02.2024
Information of production	G. Luschtinetz	23.02.2024	23.02.2024
Quality alert	G. Luschtinetz	23.02.2024	23.02.2024
Check the production-, quality- and maintenance data	G. Luschtinetz	23.02.2024	23.02.2024
Confirmation of sorting parts at customer (51.292 pcs/ cost 1.070 eur & 30.207 pcs/ 670 eu	G. Luschtinetz	23.02.2024	23.02.2024

8D report

PLAN - DO (D4-D6)

D 4. Root cause of occurrence

Work out the root causes using Ishikawa and 5 why's

Root cause of occurrence :

The guiding rail of the counter holder cylinder (the cylinder opposite of the welding cylinder) was worn. The counter holder cylinder has to much space and did not hold the position during the forming of the welding square. The proper shape was not given and in some cases the welding flags occured. Also the distance way measurement reached not the parameter and no welding current flow released.

How was the defect reproduced ?

Usage of the worn guiding rail.

D 5. Root cause of non detection

Work out the root causes using Ishikawa and 5 why's

Root cause of non detection :

The batch was blocked after the detection of the worn guiding rail. The sorting was not done. The responsiblity was not clearly defined.

D 6. Corrective actions and non detection

Actions	Resp.	Date plan.	Date clos.
When guiding elements are worn or changed (welding machine) the current batch must be blocked by the maintenance employees ==> Add the requirement to block batches in the work instruction	Daniel Beil	15.03.2024	15.03.2024
Training of the maintenance employees	Gabriel Dohr	28.03.2024	04.10.2024
FMEA update	FMEA Team	11.04.2024	11.04.2024
Guiding rail changed	Gabriel Dohr	28.03.2024	04.10.2024

CHECK - ACT (D7-D8)

D 7. Effectiveness of action plan

Actions to check effectiveness (to plan with part D6)	Resp.	Date plan.	Date clos.
Workplace audit -especially interview the employees	Audit team	09.04.2024	11.04.2024

D 8. Lessons learned

8.1 Do following documents require an update ?

	Resp.	Date plan.	Date clos.	
PFMEA	Team	11.04.2024	11.04.2024	
		verified	changed	
			yes	no
Control Plan	Team			
		verified	changed	
			yes	**no**
Work instruction				
Control instruction				
Procedure				
Control gage				
Maintenance instruction				
Lesson Learned Card				
Other				

8.2 Generalization of countermeasures to similar lines/ products

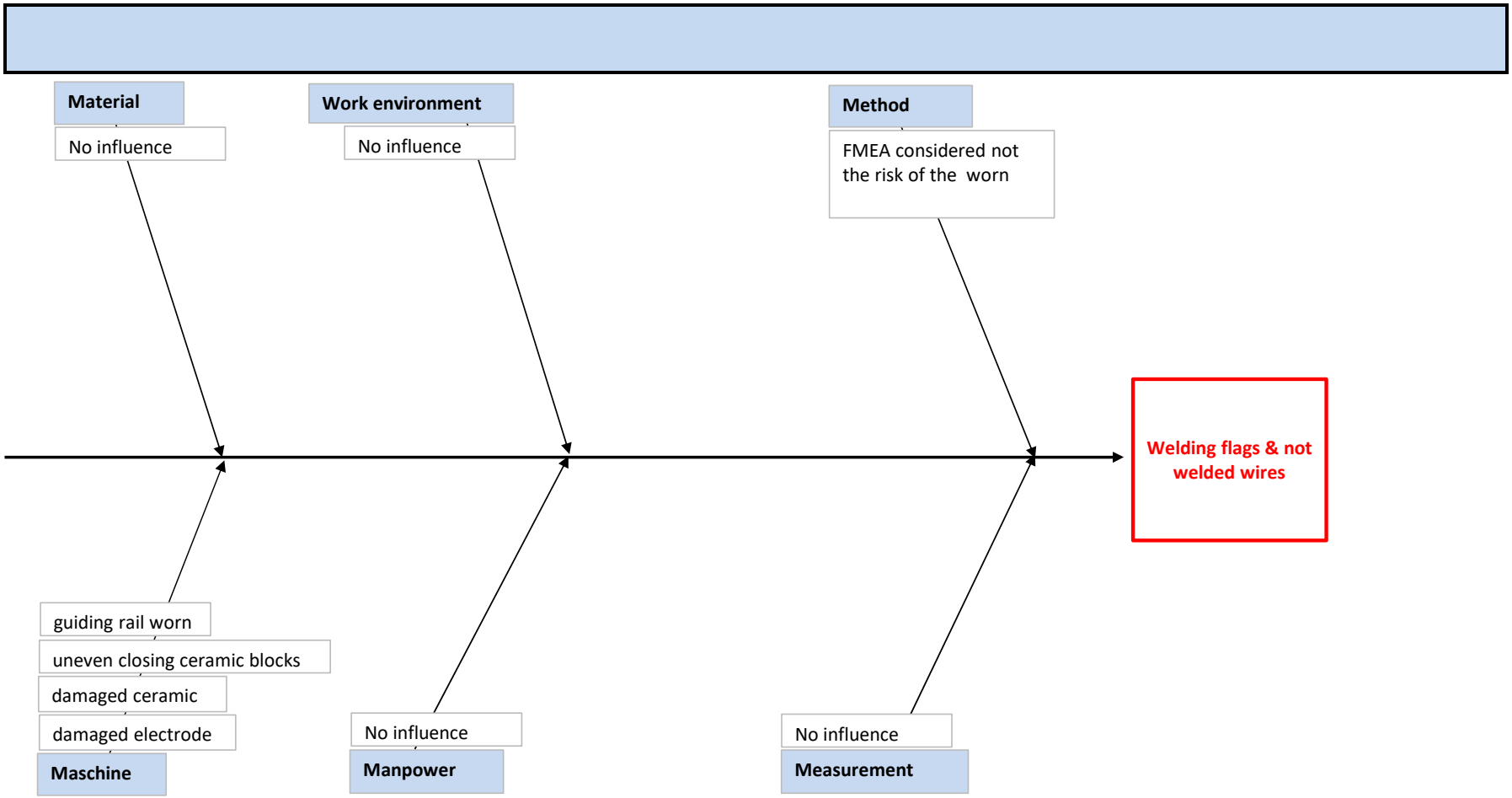
Work instruction is valid for all welding machines

8.3 Internal audit (if required)

Date 09.04.24 Results without deviation

Comments

Ishikawa


Material
No influence
Work environment
No influence
Method
FMEA considered not the risk of the worn
Welding flags & not welded wires
guiding rail worn
uneven closing ceramic blocks
damaged ceramic
damaged electrode
Maschine
No influence
Manpower
No influence
Measurement

5 Why's

Problem description :

Problem statement	Nb.	1st WHY?	1st ANSWER	2nd WHY ?	2nd ANSWER	3rd WHY ?	3rd ANSWER	4th WHY ?	4th ANSWER	5th WHY ?	5th ANSWER
OCCURENCE											
	1	Why is the shape of the welding not okay?	Because the welding square is not properly formed.	Why is the welding square not properly formed?	Because the counter holder cylinder is not straight to the other parts.	Why is the counter holder cylinder not straight to the other parts?	Because the position changend during the forming.	Why changed the position during the forming?	Because the guiding rail is worn and it is too much space between guiding and barrel bearing.	Why is the guiding rail worn and is it too much space between guiding and barrel bearing?	Because the guiding rail was not changed.
		6th WHY?	**6th ANSWER**	**7th WHY ?**	**7th ANSWER**	**8th WHY ?**	**8th ANSWER**		=> TRC		
		Why was the guiding rail not changed?	Because no spare parts were available.	Why were no spare part available?	Because no minimum quantity was defined.	Why was no minimum quantity defined?	Because the risk was not considered in the FMEA.				
					=>MRC		=>MRC				
NON DETECTION											
	1	Why was the defect not detected?	Because the batch was not sorted.	Why was the batch not sorted?	Because the batch was not blocked.	Why was the batch not blocked?	Because who block the batch was not clearly defined.				